Identification of an Unknown: Sample A

Terra Black

11-30-23

CHEM 2081-007

Introduction

Being able to identify an unknown compound is an important skill not only in chemistry but also in other applications of life. Using information such as the number of carbon-hydrogen bonds, the molecular weight of compounds, and the types of bonds present in the compound can be useful in determining the identity of an unknown compound. This information is presented in a sample report in the form of an NMR, a GCMS, and an FTIR. After the unknown compound is identified, it can be used to understand interactions between similar and different compounds. The critical thinking skills developed during the deciphering of the information in these three spectrums are an important life skill that can be applied outside the classroom and used in almost any setting of life.

Discussion

The three types of spectroscopies used in this experiment are Nuclear Magnetic Resonance (NMR), Gas Chromatography and Mass Spectrometry (GCMS), and Fourier Transform Infrared (FTIR). Each of them plays a different role in determining the identity of the unknown compound in this experiment.

NMR or Nuclear Magnetic Resonance Spectroscopy is an instrument that is used to identify the number of protons and general structure of a compound. The first NMR type used signals detected by a match between radio frequency energy and "the energy difference between two spin states of the nucleus" to identify the number of protons in the nucleus of the compound.[1] Today, the "pulsed Fourier transform NMR (FT NMR)" is the more commonly used technique. It uses a "broad pulse of electromagnetic radiation" to excite the carbon nuclei which results in a "continually decreasing oscillation"[1]. This differs from the first method because it uses carbon rather than protons. The variation in magnitude around the nucleus that is collected creates a curve which is called a "free-induction decay (FID)"[1] that is then converted into frequencies. A computer program is used to detect the multiple pulses or signals from the machine and collect data from them that can then be used to make an NMR spectrum.

NMR Spectrometers use a high magnetic field to collect data and the magnets used are highly activated, so they are kept in "double-jacketed Dewar vessels cooled by liquid helium and liquid nitrogen"[1]. The magnet used in NMR spectroscopy is 200 to 400 MHz, meaning it is very strong and caution should be taken when handling this instrument. The benefits to using NMR which has a high frequency and high field strength is that it has greater sensitivity than other instruments allowing it to have stronger sample signals when compared to "background noise", and it has greater signal separation which allows for easier interpretation of the data. There are four different types of information that can be deciphered from an NMR spectrum. These include, the number of different protons, integrations, positions of the signals, and spin-spin coupling patterns. All this information can be determined using the number of signals and the size of each signal. The number of peaks or integration lines in each signal can be used to identify the number of carbon-hydrogen bonds in a compound which helps when drawing the general structure of a compound. Although it gives a lot of information about the unknown, other spectrum types are needed to completely identify the compound.

The next type of spectroscopy that is used in this experiment is Gas Chromatography and Mass Spectrometry or GCMS. Gas Chromatography and Mass Spectrometry are two different instruments that work together to give information about the number and possible identities of the components of a compound mixture[2]. These two instruments are connected and the effluent stream from the gas chromatograph flows into the mass spectrometer. The molecules that entered the spectrometer are then passed into the mass analyzer after undergoing a series of steps. First, the ions that entered the spectrometer are hit with a stream of electrons with 70 electron volts of energy[2]. This separates the molecules based on charge. The negative ions are attracted to an electron trap which removes them from the chamber, while the positive ions are "propelled toward the mass analyzer by the positively charged repeller plate"[2]. Once in the mass analyzer the molecules undergo ionization, and both charged and uncharged fragments of the molecule are formed. The uncharged molecules and fragments are removed by a vacuum and only the charged ions are analyzed by their reaction to electric and magnetic fields[2].

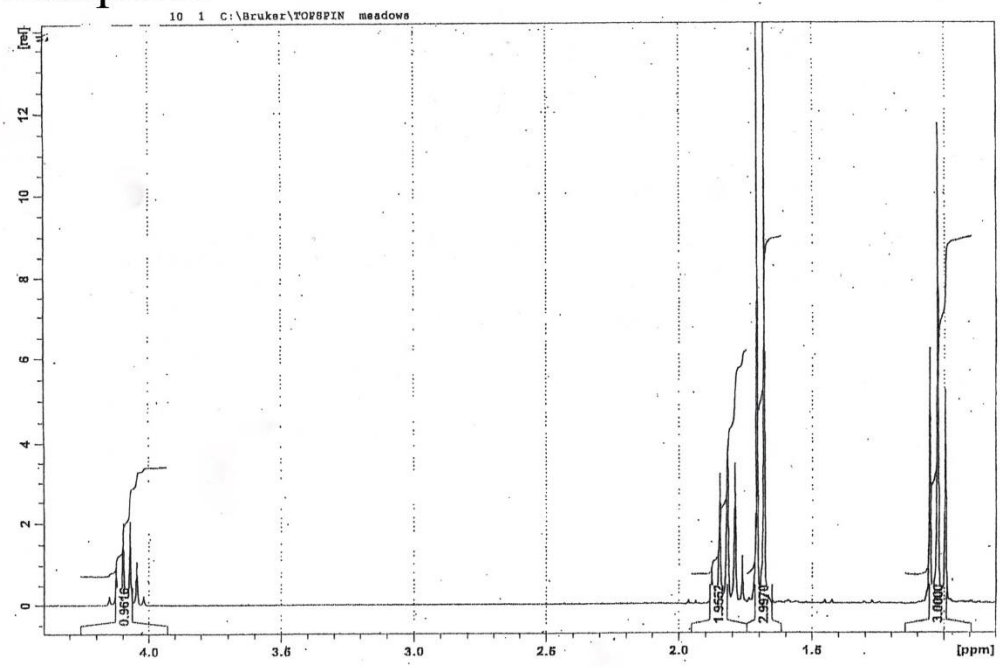
The GCMS instrument uses a quadruple mass filter which consists of four parallel stainless-steel rods. The rod pairs each have opposing direct current (DC) voltages that are superimposed with alternating current (AC) voltages. The ions are separated based on their ability or inability to reach the detector in the mass analyzer. The AC and DC voltages and frequencies determine the different readings of the ions. The data collected from the GCMS helps in identifying the major components of a compound based on the *m/z* of a molecular ion. One thing to keep in mind when using a GCMS instrument is that "most elements occur in nature as mixtures of isotopes"[2]. Depending on the intensity of the peaks relative to one another the molecular formulas for certain organic compounds can be observed. To determine the mass of a certain electronegative ion, when looking at a mass spectrometry chart, the base peak and the highest prominent peak must be observed. The base peak is the biggest and most prominent peak, while the highest prominent peak, or parent ion, is the highest peak relative to the upper end of the chart. Taking these two numbers and then subtracting them from one another will give a number that is close to the molar mass of the unknown electronegative ion. The main job of GCMS is to identify what molecules besides carbon and hydrogen are in the compound and how many molecules there are. Once this information is obtained, NMR and FTIR can be used to determine more of the structure and arrangement of the compound.

The last type of spectroscopy that was used in this experiment was Fourier Transform Infrared or FTIR. Infrared spectroscopy used in tandem with NMR is a useful tool for determining the structure of an organic molecule. It's an instrument that identifies simple functional groups in a molecule by using infrared energy that generates absorption bands when passed through a sample. When reading an FTIR the region of importance is between 4000 to 600 cm^{-1} because it is the region where absorptions of typical organic compounds most often appear. The absorptions of the molecules are measured by the vibration that occurs when the infrared radiation bends or stretches the covalent bonds that hold the molecules together in the compound. This is why there are different regions that code for different functional groups on an infrared spectrum. The two properties that are presented on infrared spectra are the wavenumber and the peak. The wavenumber is "proportional to the frequency as well as the energy of an IR absorption"[3]. The peak gives a maximum number of absorptions in wavenumber units. Not every peak on an IR absorption spectrum is relevant to the molecule, but certain wavelengths correspond to specific functional groups.

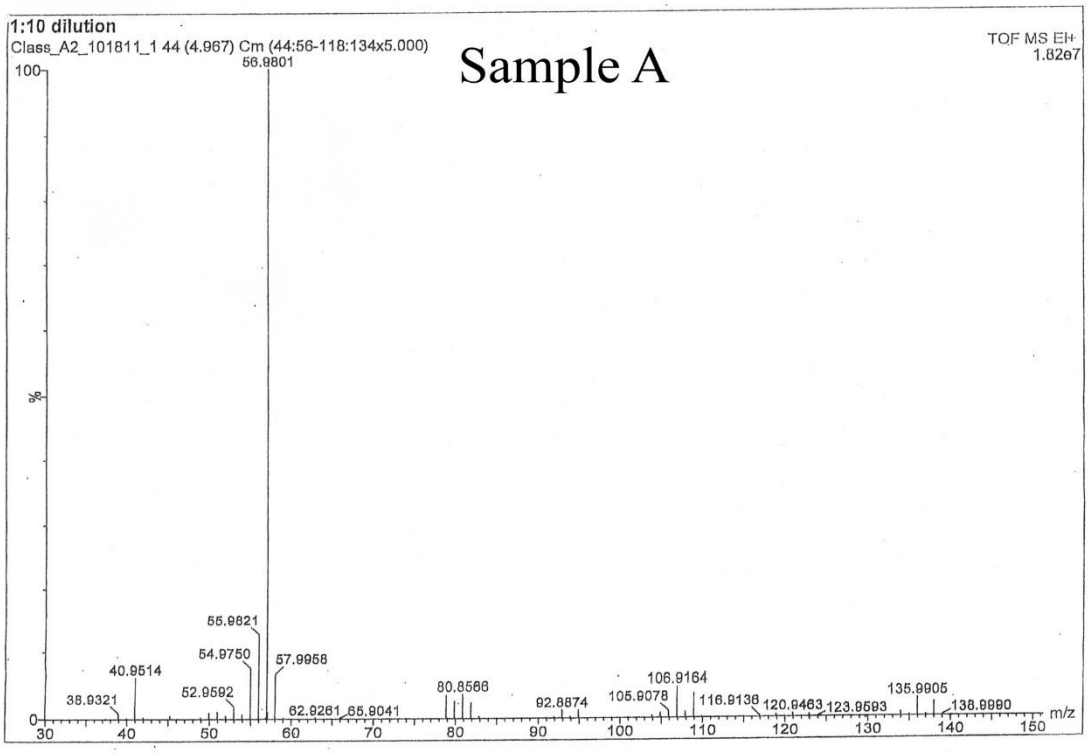
Functional groups and their respective wavelengths and regions will always be the same, so FTIR is an indispensable tool in determining the functional groups that are in an unknown compound. There are important factors that help determine the frequency and intensity of the IR peaks. The first one is the type of vibration or how the molecule stretches and bends while in a covalent bond with another molecule. Stretching vibrations will appear at a higher frequency than bending vibrations because it takes more energy to stretch covalent bonds. The second factor is the strength of the bond connecting the atoms and what order the bonds are in. Bond order is defined as the amount of bonds between each atom, and the higher the bond order the higher the frequency. This is due to the greater amount of energy that is required to stretch more than one bond. The third factor is the masses of the atoms, and the fourth factor is the electronegativity difference between the atoms in the bond[3]. When an atom is bonded to hydrogen it will have a higher frequency when compared to being bonded to a heavier compound like nitrogen. Transmittance "is the ratio of the amount of infrared radiation transmitted by the sample to the intensity of the incident beam"[3]. When this information is coupled with the information from NMR and GCMS the unknown compound's structure and molecules can be put together into a chemical compound.

NMR Sample A

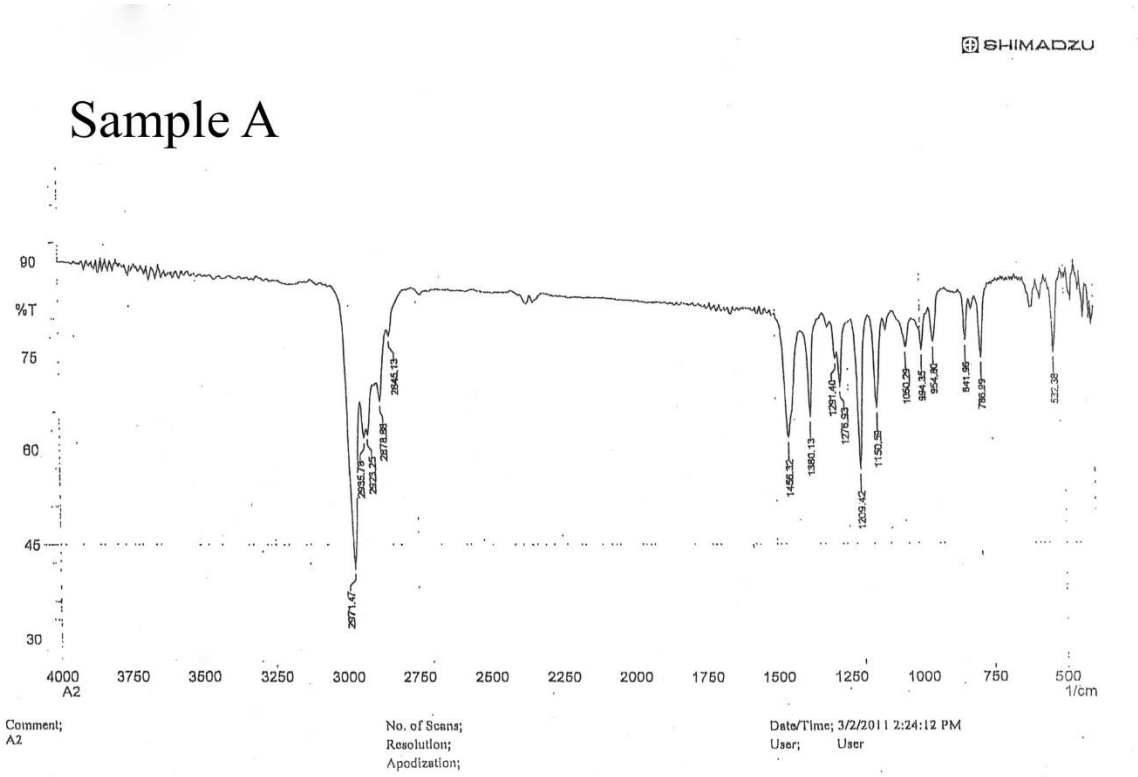
Sample A



GCMS Sample A



FTIR Sample A

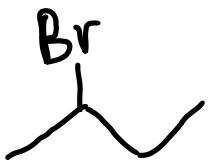


Analysis

To determine the identity of the compound, information from each chart was taken and pieced together to form the molecules that make up the compound. The NMR spectrum gives the most information about the compound, while the other two spectra help connect the dots. Looking at the NMR chart there are four peaks meaning there are four hydrogen regions in the compound, at each peak there are different numbers of integration lines. These lines show that there are two molecules that have three hydrogens, one molecule that has two hydrogens, and one molecule that has one hydrogen. The first peak, reading from right to left, has three lines which means it corresponds to two hydrogens, the second peak has two lines which means it corresponds to one hydrogen, and the third and fourth peaks have four lines which means it corresponds to three hydrogens. Now that the general structure of the compound is determined, the possible addition of any electronegative ions needs to be determined using GCMS. In GCMS, as mentioned before, the two peaks needed are the base peak and the parent ion, these are 56.9801 and 135.9905 respectively. When these two numbers are subtracted from one another the answer is approximately 79 which is close to the molar mass of bromine, 79.904. Therefore, the compound has a bromine atom attached to it. To determine how these atoms are all connected the FTIR spectrum is used. There is only one prominent peak and functional group region that is shown on this chart, which is the peak that corresponds to a C-H bond in the region between 3600-2850. The C-H peak is sharper than an O-H or N-H peak in this region, and there are no other peaks in the diagnostic region of the spectrum. This means that the compound is an alkane. The only information left to be determined is where the bromine goes on the compound. Looking back at the NMR, the bromine can’t go on either end of the molecule because the peaks that correspond with one hydrogen is closest to the electronegative ion, and there needs to be two areas of the compound that have three hydrogens. This means the bromine must be on one of the middle carbons. It cannot be on the third carbon because it must be connected to two other carbons and have one hydrogen. This leaves the second carbon. Putting all this information together leads to the name of our compound which is 2-bromobutane.

Conclusion

The unknown compound for sample A is 2-bromobutane which was found using different data points from each spectrum. The molecular formula for the compound is CH3CHBrCH2CH3.



References

1. Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University. (2023). Appendix II: NMR Instrumentation, pp. 85-132.
2. Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University. (2023). Appendix IV: Mass Spectrometers, pp. 157-174
3. Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University. (2023). Appendix III: FTIR. In CHEM 2081 Organic Chemistry II Lab Manual, pp. 133-156.